Ex.NO:04

Date : 30/07/2026

Setup and configure a LAN (Local Area network) using a switch & Ethernet cables in your lab.

Aim :

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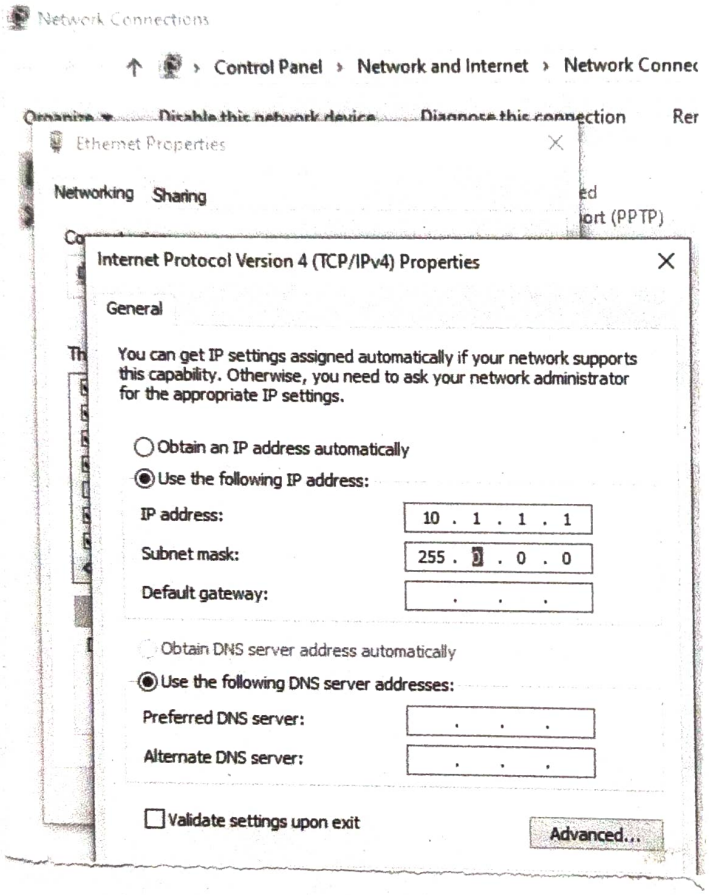
What is a LAN ?

A Local Area Network (LAN) refers to a network that connects devices within a limited area, such as an office building, school, or home. It enables users to share resources, including data, printers, and internet access. LAN connects devices to promote collaboration & transfer information between users, such as computers, printers, servers, and switches.

How to set up a LAN :

*. Step 1 : Plan and design an appropriate network topology taking into account network requirements & equipment location.

*. Step 2 : you can take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes, and 4


Network Connections
Control Panel › Network and Internet › Network Connec
Ethernet Properties
Networking Sharing
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 10 . 1 . 1 . 1
Subnet mask: 255 . 0 . 0 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...

Ethernet cables.

* Step 3 : Connect your computers to network switch via an ethernet cable, which is as simple as plugging one end of the ethernet cable into your computer & the other end into your network switch.

* Step 4 : Assign IP address to your PCs

1. Log on to the client computer as Administrator or as owner.
2. Click Network & Internet Connections.
3. Right click Local Area Connection / Ethernet -> Go to properties -> select internet protocol (TCP / IPv4) -> click on properties -> select use the following ip address option & assign ip address.

* Step 5 : configure a network switch

1. Connect your computer to the switch.
2. Log in to the web interface
3. Configure the basic settings
4. Assign IP address as : 10.1.1.5 subnet mark 255.0.0.

* Step 6 : Check the connectivity between switch and other machine by using ping command in the command prompt of the device.

*. Step 7 : Select folder, -> go to properties -> click sharing tab -> share it with everyone on the same LAN.

*. Step 8 : Try to access the shared folder from others computers of the network.

Result :

Thus the setup and configure or LAN using switch and ethernet cable was successfully completed.